

# **Text Classification**

The Task of Text Classification

# Positive or negative movie review?



- unbelievably disappointing



- Full of zany characters and richly applied satire, and some great plot twists



- this is the greatest comedy ever filmed



- It was pathetic. The worst part about it was the boxing scenes.

# Text Classification

- Assigning subject categories or topics
- Spam detection
- Authorship identification
- Age/gender identification
- Language Identification
- Sentiment analysis
- ...

# Text Classification: definition

- *Input:*
  - a document  $d$
  - a fixed set of classes  $C = \{c_1, c_2, \dots, c_J\}$
- *Output:* a predicted class  $c \in C$

# **There is mainly three text classification approach**

Rule-based System

Machine System

Hybrid System

# Classification Methods:

## Hand-coded rules

- Rules based on combinations of words or other features
  - spam: black-list-address OR (“dollars” AND “have been selected”)
- Accuracy can be high
  - If rules carefully refined by expert
- But building and maintaining these rules is expensive

# Classification Methods:

## Supervised Machine Learning

- *Input:*
  - a document  $d$
  - a fixed set of classes  $C = \{c_1, c_2, \dots, c_J\}$
  - A training set of  $m$  hand-labeled documents  $(d_1, c_1), \dots, (d_m, c_m)$
- *Output:*
  - a learned classifier  $\gamma: d \rightarrow c$

# Classification Methods:

## Supervised Machine Learning

- Any kind of classifier
  - Naïve Bayes
  - Logistic regression
  - Support-vector machines
  - k-Nearest Neighbors
  - ...



# Naïve Bayes Intuition

- Simple (“naïve”) classification method based on Bayes rule
- Relies on very simple representation of document
  - Bag of words

# The bag of words representation

Y (

I **love** this movie! It's **sweet**, but with **satirical** humor. The dialogue is **great** and the adventure scenes are **fun**... It manages to be **whimsical** and **romantic** while **laughing** at the conventions of the fairy tale genre. I would **recommend** it to just about anyone. I've seen it **several** times, and I'm always **happy** to see it **again** whenever I have a friend who hasn't seen it yet.

) = C



# The bag of words representation: using a subset of words

$$Y(\text{love sweet satirical great fun whimsical romantic laughing recommend several happy again}) = C$$

thumbs up

thumbs down

# The bag of words representation

$Y($

|           |     |
|-----------|-----|
| great     | 2   |
| love      | 2   |
| recommend | 1   |
| laugh     | 1   |
| happy     | 1   |
| ...       | ... |

$) = C$

  


# Bayes' Rule Applied to Documents and Classes

- For a document  $d$  and a class  $c$

$$P(c | d) = \frac{P(d | c)P(c)}{P(d)}$$

# Naïve Bayes Classifier (I)

$$c_{MAP} = \operatorname{argmax}_{c \in C} P(c | d)$$

MAP is “maximum a posteriori” = most likely class

$$= \operatorname{argmax}_{c \in C} \frac{P(d | c)P(c)}{P(d)}$$

Bayes Rule

$$= \operatorname{argmax}_{c \in C} P(d | c)P(c)$$

Dropping the denominator

## Naïve Bayes Classifier (II)

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(d | c)P(c)$$

$$= \underset{c \in C}{\operatorname{argmax}} P(x_1, x_2, \dots, x_n | c)P(c)$$

Document  $d$   
represented as  
features  
 $x_1 \dots x_n$

# Multinomial Naïve Bayes Independence Assumptions

$$P(x_1, x_2, \dots, x_n \mid c)$$

- **Bag of Words assumption:** Assume position doesn't matter
- **Conditional Independence:** Assume the feature probabilities  $P(x_i \mid c_j)$  are independent given the class  $c$ .

$$P(x_1, \dots, x_n \mid c) = P(x_1 \mid c) \cdot P(x_2 \mid c) \cdot P(x_3 \mid c) \cdot \dots \cdot P(x_n \mid c)$$



# Multinomial Naïve Bayes Classifier

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(x_1, x_2, \dots, x_n | c) P(c)$$

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}} P(c_j) \prod_{x \in X} P(x | c)$$

# Learning the Multinomial Naïve Bayes Model

- First attempt: maximum likelihood estimates
  - simply use the frequencies in the data

$$\hat{P}(c_j) = \frac{\text{doccount}(C = c_j)}{N_{doc}}$$

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$

# Parameter estimation

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$

fraction of times word  $w_i$  appears  
among all words in documents of topic  $c_j$

- Create mega-document for topic  $j$  by concatenating all docs in this topic
  - Use frequency of  $w$  in mega-document

# Problem with Maximum Likelihood

- What if we have seen no training documents with the word *fantastic* and classified in the topic **positive** (*thumbs-up*)?

$$\hat{P}(\text{"fantastic"} \mid \text{positive}) = \frac{\text{count}(\text{"fantastic"}, \text{positive})}{\sum_{w \in V} \text{count}(w, \text{positive})} = 0$$

# Laplace (add-1) smoothing for Naïve Bayes

$$\hat{P}(w_i | c) = \frac{\text{count}(w_i, c) + 1}{\sum_{w \in V} (\text{count}(w, c) + 1)}$$

$$= \frac{\text{count}(w_i, c) + 1}{\sum_{w \in V} \text{count}(w, c) + |V|}$$

# Multinomial Naïve Bayes: Learning

- From training corpus, extract *Vocabulary*
- Calculate  $P(c_j)$  terms
  - For each  $c_j$  in  $C$  do
    - $docs_j \leftarrow$  all docs with class =  $c_j$
$$P(c_j) \propto \frac{|docs_j|}{|\text{total \# documents}|}$$
- Calculate  $P(w_k | c_j)$  terms
  - $Text_j \leftarrow$  single doc containing all  $docs_j$
  - For each word  $w_k$  in *Vocabulary*
    - $n_k \leftarrow$  # of occurrences of  $w_k$  in  $Text_j$
$$P(w_k | c_j) \propto \frac{n_k + a}{n + a | \text{Vocabulary} |}$$

# Text Classification Example

We have an existing set of documents (D1-D5) that are categorized into Auto, Sports, and Computer.

| Document # | Content             | Category |
|------------|---------------------|----------|
| D1         | Saturn Dealer's Car | Auto     |
| D2         | Toyota Car Tercel   | Auto     |
| D3         | Baseball Game Play  | Sports   |
| D4         | Pulled Muscle Game  | Sports   |
| D5         | Colored GIFs Root   | Computer |

# Text Classification Example

Now the task is to categorize the new D6 and D7 into Auto, Sports, or Computer.

| Document # | Content              | Category |
|------------|----------------------|----------|
| D6         | Home Runs<br>Game    | ?        |
| D7         | Car Engine<br>Noises | ?        |

In machine learning, the given set of documents used to train the probabilistic model is called the training set.



# Text Classification Example

## Applying Multinomial Bayes Classification

**Step 1:** Calculate prior probabilities. These are the probability of a document being in a specific category from the given set of documents.

$P(\text{Category}) = (\text{No. of documents classified into the category}) \text{ divided by } (\text{Total number of documents})$

$P(\text{Auto}) = (\text{No of documents classified into Auto}) \text{ divided by } (\text{Total number of documents})$   
 $= 2/5 = 0.4$

$P(\text{Sports}) = 2/5 = 0.4$

$P(\text{Computer}) = 1/5 = 0.2$

# Text Classification Example

**Step 2:** Calculate Likelihood. Likelihood is the conditional probability of a word occurring in a document given that the document belongs to a particular category.

$P(\text{Word/Category}) = (\text{Number of occurrence of the word in all the documents from a category} + 1) \text{ divided by } (\text{All the words in every document from a category} + \text{Total number of unique words in all the documents})$

$P(\text{Saturn/Auto}) = (\text{Number of occurrence of the word "SATURN" in all the documents in "AUTO"} + 1) \text{ divided by } (\text{All the words in every document from "AUTO"} + \text{Total number of unique words in all the documents})$

$$= (1+1)/(6+13) = 2/19 = 0.105263158$$

# Auto

| Word     | # of Occurrences of Word in Auto | Total Words in Auto | Conditional Probability of Given Word in Auto | # of Total Unique Words in All Documents |
|----------|----------------------------------|---------------------|-----------------------------------------------|------------------------------------------|
| Saturn   | 1                                | 6                   | 0.105263158                                   | 13                                       |
| Dealers  | 1                                | 6                   | 0.105263158                                   | 13                                       |
| Car      | 2                                | 6                   | 0.157894737                                   | 13                                       |
| Toyota   | 1                                | 6                   | 0.105263158                                   | 13                                       |
| Tercel   | 1                                | 6                   | 0.105263158                                   | 13                                       |
| Baseball | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Game     | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Play     | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Pulled   | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Muscle   | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Colored  | 0                                | 6                   | 0.052631579                                   | 13                                       |
| GIFs     | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Root     | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Home     | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Runs     | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Engine   | 0                                | 6                   | 0.052631579                                   | 13                                       |
| Noises   | 0                                | 6                   | 0.052631579                                   | 13                                       |

# Sports

| Word     | # of Occurrences of Word in Sports | Total Words in Sports | Conditional Probability of Given Word | # of Total Unique Words in All Documents |
|----------|------------------------------------|-----------------------|---------------------------------------|------------------------------------------|
| Saturn   | 0                                  | 6                     | 0.052631579                           | 13                                       |
| Dealers  | 0                                  | 6                     | 0.052631579                           | 13                                       |
| Car      | 0                                  | 6                     | 0.052631579                           | 13                                       |
| Toyota   | 0                                  | 6                     | 0.052631579                           | 13                                       |
| Tercel   | 0                                  | 6                     | 0.052631579                           | 13                                       |
| Baseball | 1                                  | 6                     | 0.105263158                           | 13                                       |
| Game     | 2                                  | 6                     | 0.157894737                           | 13                                       |
| Play     | 1                                  | 6                     | 0.105263158                           | 13                                       |
| Pulled   | 1                                  | 6                     | 0.105263158                           | 13                                       |
| Muscle   | 1                                  | 6                     | 0.105263158                           | 13                                       |
| Colored  | 1                                  | 6                     | 0.105263158                           | 13                                       |
| GIFs     | 1                                  | 6                     | 0.105263158                           | 13                                       |
| Root     | 1                                  | 6                     | 0.105263158                           | 13                                       |
| Home     | 0                                  | 6                     | 0.052631579                           | 13                                       |
| Runs     | 0                                  | 6                     | 0.052631579                           | 13                                       |
| Engine   | 0                                  | 6                     | 0.052631579                           | 13                                       |
| Noises   | 0                                  | 6                     | 0.052631579                           | 13                                       |

# Computer

| Word     | # of Occurrences of Word in Computer | Total Words in Computer | Conditional Probability of Given Word in Computer | # of Total Unique Words in All Documents |
|----------|--------------------------------------|-------------------------|---------------------------------------------------|------------------------------------------|
| Saturn   | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Dealers  | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Car      | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Toyota   | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Tercel   | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Baseball | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Game     | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Play     | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Pulled   | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Muscle   | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Colored  | 1                                    | 3                       | 0.125                                             | 13                                       |
| GIFs     | 1                                    | 3                       | 0.125                                             | 13                                       |
| Root     | 1                                    | 3                       | 0.125                                             | 13                                       |
| Home     | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Runs     | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Engine   | 0                                    | 3                       | 0.0625                                            | 13                                       |
| Noises   | 0                                    | 3                       | 0.0625                                            | 13                                       |

### Step 3

Calculate  $P(\text{Category}/\text{Document}) = P(\text{Category}) * P(\text{Word1}/\text{Category}) * P(\text{Word2}/\text{Category}) * P(\text{Word3}/\text{Category})$

$$\begin{aligned} P(\text{Auto}/\text{D6}) &= P(\text{Auto}) * P(\text{Home}/\text{Auto}) * P(\text{Runs}/\text{Auto}) * P(\text{Game}/\text{Auto}) \\ &= (0.4) * (0.052631579) * (0.052631579) * (0.052631579) \\ &= (0.00005831754) \end{aligned}$$

$$P(\text{Sports}/\text{D6}) = 0.000174953$$

$$P(\text{Computers}/\text{D6}) = 0.00004882813$$

The most probable category for **D6 to fall into is Sports**, because it has the highest probability among its peers.

### Step 3

Calculate  $P(\text{Category}/\text{Document}) = P(\text{Category}) * P(\text{Word1}/\text{Category}) * P(\text{Word2}/\text{Category}) * P(\text{Word3}/\text{Category})$

$$\begin{aligned} P(\text{Auto}/D7) &= P(\text{Auto}) * P(\text{Engine}/\text{Auto}) * P(\text{Noises}/\text{Auto}) * P(\text{Car}/\text{Auto}) \\ &= (0.4) * (0.052631579) * (0.052631579) * (0.157894737) \\ &= 0.00017495262 \end{aligned}$$

$$P(\text{Sports}/D7) = 0.0000583175$$

$$P(\text{Computers}/D7) = 0.00004882813$$

The most probable category for **D7 to fall into is Auto**, because it has the highest probability among its peers.

# Laplace (add-1) smoothing: unknown words

Add one extra word to the vocabulary, the “unknown word”  $w_u$

$$\hat{P}(w_u | c) = \frac{\text{count}(w_u, c) + 1}{\sum_{w \in V} \text{count}(w, c) + |V + 1|}$$

$$= \frac{1}{\sum_{w \in V} \text{count}(w, c) + |V + 1|}$$



# Naïve Bayes and Language Modeling

- Naïve bayes classifiers can use any sort of feature
  - URL, email address, dictionaries, network features
- But if, as in the previous slides
  - We use **only** word features
  - we use **all** of the words in the text (not a subset)
- Then
  - Naïve bayes has an important similarity to language modeling.

# Each class = a unigram language model

- Assigning each word:  $P(\text{word} \mid c)$
- Assigning each sentence:  $P(s \mid c) = \prod P(\text{word} \mid c)$

Class *pos*

|      |      |          |             |             |            |             |
|------|------|----------|-------------|-------------|------------|-------------|
| 0.1  | I    | <u>I</u> | <u>love</u> | <u>this</u> | <u>fun</u> | <u>film</u> |
| 0.1  | love | 0.1      | 0.1         | .05         | 0.01       | 0.1         |
| 0.01 | this |          |             |             |            |             |
| 0.05 | fun  |          |             |             |            |             |
| 0.1  | film |          |             |             |            |             |

$$P(s \mid \text{pos}) = 0.00000005$$

# Naïve Bayes as a Language Model

- Which class assigns the higher probability to s?

| Model pos |      |
|-----------|------|
| 0.1       | I    |
| 0.1       | love |
| 0.01      | this |
| 0.05      | fun  |
| 0.1       | film |

| Model neg |      |
|-----------|------|
| 0.2       | I    |
| 0.001     | love |
| 0.01      | this |
| 0.005     | fun  |
| 0.1       | film |

| <u>I</u> | <u>love</u> | <u>this</u> | <u>fun</u> | <u>film</u> |
|----------|-------------|-------------|------------|-------------|
| 0.1      | 0.1         | 0.01        | 0.05       | 0.1         |
| 0.2      | 0.001       | 0.01        | 0.005      | 0.1         |

$$P(s|\text{pos}) > P(s|\text{neg})$$

# The 2-by-2 contingency table

|              | correct | not correct |
|--------------|---------|-------------|
| selected     | tp      | fp          |
| not selected | fn      | tn          |

# Precision and recall

- **Precision:** % of selected items that are correct  
**Recall:** % of correct items that are selected

|              | correct | not correct |
|--------------|---------|-------------|
| selected     | tp      | fp          |
| not selected | fn      | tn          |

# A combined measure: F

- A combined measure that assesses the P/R tradeoff is F measure (weighted harmonic mean):
- $F = 2PR/(P+R)$

# Sentiment Analysis

# What is Sentiment Analysis?

The key aspect of sentiment analysis is to analyze a body of text for understanding the opinion expressed by it. Typically, we quantify this sentiment with a positive or negative value, **called polarity**. The overall sentiment is often inferred as positive, neutral or negative from the sign of the polarity score.



# How is Sentiment Analysis Used?

Usually, sentiment analysis works best on text that has a **subjective context** than on text with only an objective context. **Objective text** usually depicts some normal statements or facts without expressing any emotion, feelings, or mood.

Subjective text contains text that is usually expressed by a human having typical moods, emotions, and feelings. Sentiment analysis is widely used, especially as a part of social media analysis for any domain, be it a business, a recent movie, or a product launch, to understand its reception by the people and what they think of it based on their opinions or, you guessed it, sentiment!

# **Sentiment analysis has many other names**

- Opinion extraction
- Opinion mining
- Sentiment mining
- Subjectivity analysis

# Lexicons used for sentiment analysis

A lexicon is a dictionary, vocabulary, or a book of words. In our case, lexicons are special dictionaries or vocabularies that have been created for analyzing sentiments.

Most of these lexicons have a list of positive and negative polar words with some score associated with them, and using various techniques like the position of words, surrounding words, context, parts of speech, phrases, and so on, scores are assigned to the text documents for which we want to compute the sentiment. After aggregating these scores, we get the final sentiment.

Various popular lexicons are used for sentiment analysis, including the following.

- AFINN lexicon
- Bing Liu's lexicon
- MPQA subjectivity lexicon
- SentiWordNet
- VADER lexicon
- TextBlob lexicon

# Why sentiment analysis?

- *Movie*: is this review positive or negative?
- *Products*: what do people think about the new iPhone?
- *Public sentiment*: how is consumer confidence? Is despair increasing?
- *Politics*: what do people think about this candidate or issue?
- *Prediction*: predict election outcomes or market trends from sentiment

# Scherer Typology of Affective States

- **Emotion:** brief organically synchronized ... evaluation of a major event
  - *angry, sad, joyful, fearful, ashamed, proud, elated*
- **Mood:** diffuse non-caused low-intensity long-duration change in subjective feeling
  - *cheerful, gloomy, irritable, listless, depressed, buoyant*
- **Interpersonal stances:** affective stance toward another person in a specific interaction
  - *friendly, flirtatious, distant, cold, warm, supportive, contemptuous*
- **Attitudes:** enduring, affectively colored beliefs, dispositions towards objects or persons
  - *liking, loving, hating, valuing, desiring*
- **Personality traits:** stable personality dispositions and typical behavior tendencies
  - *nervous, anxious, reckless, morose, hostile, jealous*

# Scherer Typology of Affective States

- **Emotion:** brief organically synchronized ... evaluation of a major event
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  - *nervous, anxious, reckless, morose, hostile, jealous*

# Sentiment Analysis

- Sentiment analysis is the detection of **attitudes**  
“enduring, affectively colored beliefs, dispositions towards objects or persons”
  1. **Holder (source)** of attitude
  2. **Target (aspect)** of attitude
  3. **Type** of attitude
    - From a set of types
      - *Like, love, hate, value, desire, etc.*
    - Or (more commonly) simple **polarity**:
      - *positive, negative, neutral*
  4. **Text** containing the attitude
    - Sentence or entire document



# Sentiment Analysis

- Simplest task:
  - Is the attitude of this text positive or negative?
- More complex:
  - Rank the attitude of this text from 1 to 5
- Advanced:
  - Detect the target, source, or complex attitude types

# Sentiment Analysis

- Simplest task:
  - Is the attitude of this text positive or negative?
- More complex:
  - Rank the attitude of this text from 1 to 5
- Advanced:
  - Detect the target, source, or complex attitude types

# Baseline Algorithm

- Tokenization
- Feature Extraction
- Classification using different classifiers
  - Naïve Bayes
  - Logistic Regression
  - SVM

# Sentiment Tokenization Issues

- Deal with HTML and XML markup
- Twitter mark-up (names, hash tags)
- Capitalization (preserve for

words in all caps)

- Phone numbers, dates
- Emoticons
- Useful code:

## Potts emoticons

```
[<>]?           # optional hat/brow
[:;]=8]         # eyes
[\-o\*\'\']?   # optional nose
[\)\)\)\(\[dDpP/\:\}\{\@\\|\\] # mouth
|               ##### reverse orientation
[\)\)\)\(\[dDpP/\:\}\{\@\\|\\] # mouth
[\-o\*\'\']?   # optional nose
[:;]=8]         # eyes
[<>]?           # optional hat/brow
```

# Extracting Features for Sentiment Classification

- How to handle negation
  - I **didn't** like this movie
  - vs
  - I really like this movie
- Which words to use?
  - Only adjectives
  - All words
    - All words turns out to work better, at least on this data

# Naïve Bayes

$$c_{NB} = \underset{c_j \in C}{\operatorname{argmax}} P(c_j) \prod_{i \in \text{positions}} P(w_i | c_j)$$

$$\hat{P}(w | c) = \frac{\text{count}(w, c) + 1}{\text{count}(c) + |V|}$$

# Boolean Multinomial Naïve Bayes: Learning

- From training corpus, extract *Vocabulary*

- Calculate  $P(c_j)$  terms

- For each  $c_j$  in  $C$  do

$docs_j \leftarrow$  all docs with class =  $c_j$

$$P(c_j) \propto \frac{|docs_j|}{|\text{total \# documents}|}$$

- Calculate  $P(w_k | c_j)$  terms

- Remove duplicates in each doc:

- For each word type  $w$  in  $doc_j$

- Retain only a single instance of  $w$

- $Text_j \leftarrow$  single doc containing all  $docs_j$

- For each word  $w_k$  in *Vocabulary*

$n_k \leftarrow$  # of occurrences of  $w_k$  in  $Text_j$

$$P(w_k | c_j) \propto \frac{n_k + a}{n + a | \text{Vocabulary} |}$$

# Boolean Multinomial Naïve Bayes on a test document $d$

- First remove all duplicate words from  $d$
- Then compute NB using the same equation:

$$c_{NB} = \underset{c_j \in C}{\operatorname{argmax}} P(c_j) \prod_{i \in \text{positions}} P(w_i | c_j)$$



# Thwarted Expectations

- “This film should be brilliant. It sounds like a great plot, the actors are first grade, and the supporting cast is good as well, and Stallone is attempting to deliver a good performance. However, it can’t hold up.”
- Well as usual Keanu Reeves is nothing special, but surprisingly, the very talented Laurence Fishbourne is not so good either, I was surprised.

# Using WordNet to learn polarity

- WordNet: online thesaurus (covered in later lecture).
- Create positive (“good”) and negative seed-words (“terrible”)
- Find Synonyms and Antonyms
  - Positive Set: Add synonyms of positive words (“well”) and antonyms of negative words
  - Negative Set: Add synonyms of negative words (“awful”) and antonyms of positive words (“evil”)
- It simply means emotions expressed in a sentence.
- Polarity analysis takes into account the amount of positive or negative terms that appear in a given sentence.

# Output polarity lexicon

- Positive
  - bold decisive generous good honest important large mature patient peaceful positive proud sound stimulating straightforward talented vigorous witty...
- Negative
  - ambiguous cynical evasive harmful hypocritical inefficient insecure irrational irresponsible minor reckless risky selfish tedious unsupported vulnerable wasteful...